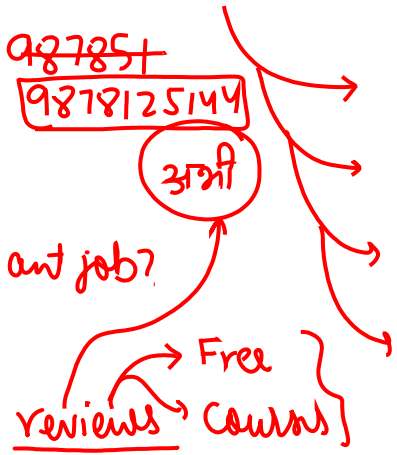
2 variables cube identities:



Pin Group

Web Version

$(a+b)^3$	$= a^3 + b^3 + 3ab(a+b)$
	$= a^3 + 3a^2b + 3ab^2 + b^3$
$(a-b)^3$	$= a^3 - b^3 - 3ab(a-b)$
	$= a^3 - 3a^2b + 3ab^2 - b^3$

$a^3 + b^3$	$= (a+b)^3 - 3ab(a+b)$
	$= (a+b)(a^2 - ab + b^2)$
$a^3 - b^3$	$= (a-b)^3 + 3ab(a-b)$
	$= (a-b)(a^2 + ab + b^2)$

If $p - 2q = 4$,then the value of $p^3 - 8q^3 - 24pq - 64$ is

a) 2 b) 0 c) 3 d) -1

If $p = 999$, then the value of $\sqrt[3]{p(p^2 + 3p + 3) + 1}$ is

a) 1000 b) 999 c) 998

d) 1002

If $p = 99$, then value of $p\left(p^2 + 3p + 3\right)$is

a) 999 b) 9999 c) 99999

d) 999999

If $p^3 + 3p^2 + 3p = 26$, then the value of $p^2 + 2p$ is:

a) 8 b) 12 c) 10 d)15

Alg.

SSC CHSL, 30 May 2022 S3

If $m = -9, n = 8$, then the value of $m^3 - 3m^2 + 3m + 3n + 3n^2 + n^3$ is

a) 271 b) -271 c) -1 d) 0

If $8x^3 - 27y^3 = (Ax + By)(Cx^2 - Dy^2 + 6xy)$, then $(A + B + C - D)$ is equal to:

a) -12 b) 12 c) 15 d) 9

If $27x^3 - 64y^3 = (Ax + By)(Cx^2 - Dy^2 + 12xy)$, then the value of $4A + B + 3C + 2D$ is:

a) 5 b) 3 c) -3 d) -4

SSC CGL T-2 2020

If $24\sqrt{3}x^3 + 5\sqrt{5}y^3 = \left(2\sqrt{3}x + \sqrt{5}y\right) \times \left(Ax^2 + Bxy + Cy^2\right)$,then what is the value of $\left(A^2 - B^2 + C^2\right)$?

a)108 b)128 c)109 d)139

If $250\sqrt{2}x^3 - 5\sqrt{5}y^3 = \left(5\sqrt{2}x - \sqrt{5}y\right)\left(Ax^2 + Bxy + Cy^2\right)$, then the value of $\left(A + C - \sqrt{10}B\right)$ is:

a) 10 b) 5 c) $5\sqrt{2}$ d) $2\sqrt{5}$

If $x^6 - 6\sqrt{6}y^6 = (x^2 + Ay^2)(x^4 + Bx^2y^2 + Cy^4)$, then what will be value of $(A^2 - B^2 + C^2)$?

a) 27 b) 42 c) 36 d) 18

If $8(x+y)^3-27(x-y)^3=(5y-x)\left(Ax^2+By^2+Cxy\right)$, then what is the value of $(A+B-C)$?

a) 36 b) 16 c) -26 d) -16

Given that $(2x+y)^3-(x+2y)^3=(x-y)\left[A\left(x^2+y^2\right)+Bxy\right]$,the value of $(2A-B)$ is:

a) 7 b) 6 c) 0 d) 1

SSC CHSL 2020

If $(x+2y)^3+(2x-y)^3=(3x+y)\left[Ax^2+By^2+Cxy\right]$, then the value of $(3A-B-C):(A+B+C)$ is:

a) $2:5$ b) $3:4$ c) $5:7$ d) $1:7$

[2

Alg (Prac)

$$\frac{775\times775\times775+225\times225\times225}{77.5\times77.5+22.5\times22.5-77.5\times22.5}$$ is equal to:

$$\frac{775\times775\times775+225\times225\times225}{}$$

a) 100 b) 10000 c) 100000 d) 10

b) (a^2+b^2-ab)

$$\frac{a^3+b^3}{a^2+b^2-ab}$$

SSC CHSL 2020

$$\frac{(253)^3+(247)^3}{25.3\times2.53-62.491+2.47\times24.7} = 50 \times 10^k,\ k = ?$$

a) 3 b) 4 c) 2 d) -3

Simplify the following expression.

$$\frac{(62 \times 62 \times 62) - 3(62 \times 62 \times 22) + 3(62 \times 22 \times 22) - (22 \times 22 \times 22)}{8 \times 8 \times 8}$$

a) 225 b) 1250 c) 125 d) 25

$\frac{5.169 \times 5.169 \times 5.169 - 64 \times (0.888)^3}{(5.169)^2 + (3.552)^2 + 4\,(0.888)(5.169)}$ is $(1+k)$, where k = _____.

0.323

0.617

0.623

0.317

If $a^2 + b^2 + ab = 0$, then $\left(a^3 - b^3\right)$ is equal to

a) 0 b) 1

c) $(a + b)^3$ d) $a^2b^4 + a^4b^2$

If $a^4 + b^4 = a^2b^2$,then $\left(a^6 + b^6\right)$ equals

a) 0 b) 1

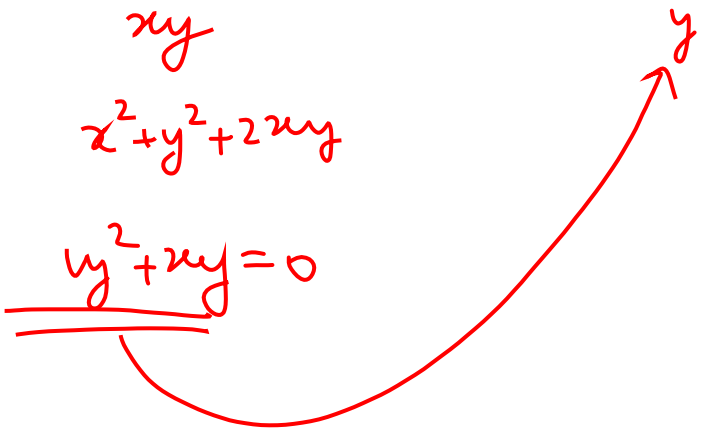
c) $a^2 + b^2$ d)$a^2b^4 + a^4b^2$

If $\frac{a}{b} + \frac{b}{a} = 1, a \neq 0, b \neq 0$ the value of $a^3 + b^3$ is

a) 0 b) 1 c) -1 d) 2

If $\frac{1}{x+y} = \frac{1}{x} + \frac{1}{y}$ then the value of $x^3 - y^3$ is

a) 0 b) 1 c) -1 d) 2



If $a^3 + 3a^2 + 9a = 1$, then what is the value of $a^3 + \left(\frac{3}{a}\right)^3$?

a) 31 b) 26 c) 28 d) 24

If $a + a^2 + a^3 - 1 = 0$, then what is the value of $a^3 + \frac{1}{a}$?

1

a) 1 b) 4 c) 2 d) 3

If $x^3 + \frac{3}{x} = 4(a^3 + b^3)$ and $3x + \frac{1}{x^3} = 4(a^3 - b^3)$, then $4(a^2 - b^2)$ is equal to:

$^3 = 4(a + b)$ $+ ^1 = 4(a - b)$ $(a - b)$

a) 0 b) 1 c) 2 d) 4